



# **FIFA 2026**

## **World Cup Database & Human Data Analytics**

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Comprehensive SQL Analysis & Power BI Visualization

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# PROJECT OVERVIEW

FIFA 2026 World Cup Database & Analytics Initiative



## Project Purpose

To conduct a comprehensive analysis of human data derived from the FIFA 2026 tournament structure, integrating diverse data points to extract meaningful insights.

### KEY DIMENSIONS

- ✓ Player Demographics
- ✓ Match Performance
- ✓ Economic Impact



## Database Schema

A relational database system comprising 6 interconnected tables designed to simulate tournament operations.

ENTITIES:  
**20 Players**

ORGANIZATIONS:  
**10 Teams**

VENUES:  
**10 Stadiums**

EVENTS:  
**12 Matches**



## Focus Areas

Strategic pillars targeted for business intelligence and operational optimization.



**Player Performance:** Skill metrics & contribution stats



**Financial Analysis:** Ticket sales & merchandise revenue



**Fan Engagement:** Demographics & spending behavior



# DATABASE SCHEMA | TABLE STRUCTURES

Comprehensive overview of 6 relational tables with data types and key constraints

- Primary Key (PK)
- Foreign Key (FK)

## TEAMS (8 Cols)

| COLUMN NAME               | TYPE    | DESCRIPTION            |
|---------------------------|---------|------------------------|
| <span>team_id (PK)</span> | INT     | Unique identifier      |
| team_name                 | VARCHAR | Official team name     |
| country_code              | CHAR    | ISO 3-letter code      |
| fifa_ranking              | INT     | Current ranking        |
| confederation             | VARCHAR | Regional confederation |
| total_world_cups_won      | INT     | Historical titles      |
| population                | BIGINT  | Country population     |
| gdp_per_capita            | DECIMAL | GDP per capita         |

## MATCHES (9 Cols)

| COLUMN NAME                    | TYPE    | DESCRIPTION       |
|--------------------------------|---------|-------------------|
| <span>match_id (PK)</span>     | INT     | Unique identifier |
| match_date                     | DATE    | Date played       |
| <span>stadium_id (FK)</span>   | INT     | Link to Stadiums  |
| <span>team_home_id (FK)</span> | INT     | Home team link    |
| <span>team_away_id (FK)</span> | INT     | Away team link    |
| goals_home                     | INT     | Home goals        |
| goals_away                     | INT     | Away goals        |
| attendance                     | INT     | Total spectators  |
| match_stage                    | VARCHAR | Tournament stage  |

## PLAYERS (10 Cols)

| COLUMN NAME                 | TYPE    | DESCRIPTION       |
|-----------------------------|---------|-------------------|
| <span>player_id (PK)</span> | INT     | Unique identifier |
| player_name                 | VARCHAR | Full name         |
| <span>team_id (FK)</span>   | INT     | Link to Teams     |
| position                    | VARCHAR | Playing position  |
| age                         | INT     | Age in years      |
| height_cm                   | INT     | Height (cm)       |
| weight_kg                   | INT     | Weight (kg)       |
| market_value_millions       | DECIMAL | Value (millions)  |
| caps                        | INT     | Total Apps        |
| goals                       | INT     | Total Goals       |

## STADIUMS (6 Cols)

| COLUMN NAME                  | TYPE    | DESCRIPTION       |
|------------------------------|---------|-------------------|
| <span>stadium_id (PK)</span> | INT     | Unique identifier |
| stadium_name                 | VARCHAR | Official name     |
| city                         | VARCHAR | Location city     |
| country                      | VARCHAR | Location country  |
| capacity                     | INT     | Seating capacity  |
| host_nation                  | VARCHAR | Host country      |

## PLAYER\_PERFORMANCE (10 Cols)

| COLUMN NAME                      | TYPE | DESCRIPTION       |
|----------------------------------|------|-------------------|
| <span>performance_id (PK)</span> | INT  | Unique identifier |
| <span>match_id (FK)</span>       | INT  | Link to Matches   |
| <span>player_id (FK)</span>      | INT  | Link to Players   |
| minutes_played                   | INT  | Mins in match     |
| goals_scored                     | INT  | Goals in match    |
| assists                          | INT  | Assists           |
| yellow_cards                     | INT  | Yellow cards      |
| red_cards                        | INT  | Red cards         |
| passes_completed                 | INT  | Successful passes |
| shots_on_target                  | INT  | Shots on target   |

## FAN\_DEMOGRAPHICS (7 Cols)

| COLUMN NAME                | TYPE    | DESCRIPTION       |
|----------------------------|---------|-------------------|
| <span>fan_id (PK)</span>   | INT     | Unique identifier |
| <span>match_id (FK)</span> | INT     | Link to Matches   |
| age_group                  | VARCHAR | Age bracket       |
| gender                     | VARCHAR | Gender ID         |
| nationality                | VARCHAR | Fan nationality   |
| ticket_price_usd           | DECIMAL | Ticket price      |
| merchandise_spent_usd      | DECIMAL | Merchandise spend |



# ENTITY RELATIONSHIP DIAGRAM

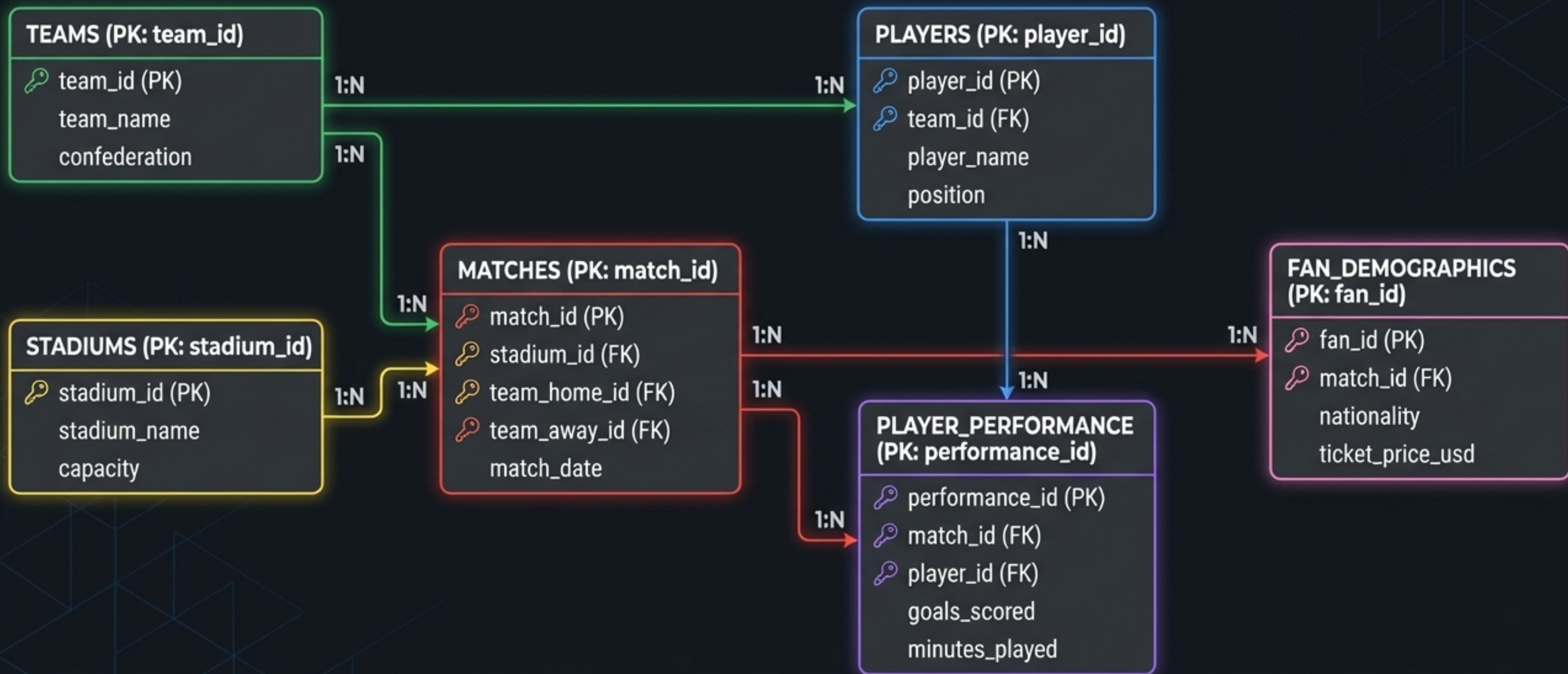
Visualizing data connectivity and cardinality constraints

🔑 PK: Primary Key

🔑 FK: Foreign Key

🔑 FK: Foreign Key

1:N: 1:Many Relationship





# SAMPLE DATA | TEAMS & PLAYERS

Populated database records used for analytics scenarios

☒ Highlighted Key Players



## TEAMS TABLE

| ID | NAME      | CODE | RANK | CONFED   | CUPS |
|----|-----------|------|------|----------|------|
| 1  | Brazil    | BRA  | 1    | CONMEBOL | 5    |
| 2  | France    | FRA  | 2    | UEFA     | 2    |
| 3  | Argentina | ARG  | 3    | CONMEBOL | 3    |
| 4  | England   | ENG  | 4    | UEFA     | 1    |
| 5  | Germany   | GER  | 5    | UEFA     | 4    |
| 6  | Spain     | ESP  | 6    | UEFA     | 1    |
| 7  | USA       | USA  | 11   | CONCACAF | 0    |
| 8  | Mexico    | MEX  | 15   | CONCACAF | 0    |
| 9  | Nigeria   | NGA  | 28   | CAF      | 0    |
| 10 | Japan     | JPN  | 17   | AFC      | 0    |



## PLAYERS TABLE

| ID | PLAYER NAME       | TEAM      | POSITION   | AGE | CAPS | VALUE (\$M) |
|----|-------------------|-----------|------------|-----|------|-------------|
| 1  | Neymar Jr         | Brazil    | Forward    | 32  | 125  | 90.00       |
| 2  | Raphinha          | Brazil    | Forward    | 27  | 22   | 65.00       |
| 3  | Kylian Mbappé     | France    | Forward    | 25  | 75   | 180.00      |
| 4  | Antoine Griezmann | France    | Midfielder | 33  | 130  | 40.00       |
| 5  | Lionel Messi      | Argentina | Forward    | 37  | 180  | 50.00       |
| 6  | Julian Alvarez    | Argentina | Forward    | 24  | 28   | 90.00       |
| 7  | Harry Kane        | England   | Forward    | 31  | 90   | 100.00      |
| 8  | Jude Bellingham   | England   | Midfielder | 21  | 29   | 180.00      |
| 9  | Joshua Kimmich    | Germany   | Midfielder | 29  | 91   | 70.00       |
| 10 | Antonio Rudiger   | Germany   | Defender   | 31  | 65   | 35.00       |
| 11 | Pedri             | Spain     | Midfielder | 22  | 22   | 100.00      |
| 12 | Christian Pulisic | USA       | Forward    | 26  | 69   | 45.00       |
| 13 | Hirving Lozano    | Mexico    | Forward    | 29  | 72   | 25.00       |
| 14 | Lamine Yamal      | Spain     | Forward    | 17  | 15   | 120.00      |
| 15 | Erling Haaland    | Japan     | Forward    | 24  | 38   | 200.00      |
| 16 | Virgil van Dijk   | England   | Defender   | 33  | 60   | 40.00       |
| 17 | Victor Osimhen    | Nigeria   | Forward    | 25  | 27   | 85.00       |
| 18 | Kaoru Mitoma      | Japan     | Forward    | 27  | 20   | 45.00       |
| 19 | Jamal Musiala     | Germany   | Midfielder | 21  | 25   | 110.00      |
| 20 | Bukayo Saka       | England   | Forward    | 22  | 35   | 140.00      |



# SAMPLE DATA I MATCHES & STADIUMS

Tournament schedule and venue information with sample results

Highlighted Final Match

## STADIUMS TABLE (10 Records)

| ID | STADIUM NAME    | CITY        | CTRY | CAP. |
|----|-----------------|-------------|------|------|
| 1  | MetLife Stadium | New York    | USA  | 82k  |
| 2  | AT&T Stadium    | Dallas      | USA  | 80k  |
| 3  | Rose Bowl       | Los Angeles | USA  | 88k  |
| 4  | Azteca Stadium  | Mexico City | MEX  | 87k  |
| 5  | BC Place        | Vancouver   | CAN  | 54k  |
| 6  | SoFi Stadium    | Los Angeles | USA  | 70k  |
| 7  | Lumen Field     | Seattle     | USA  | 69k  |
| 8  | NRG Stadium     | Houston     | USA  | 72k  |
| 9  | Mercedes-Benz   | Atlanta     | USA  | 71k  |
| 10 | BMO Field       | Toronto     | CAN  | 45k  |

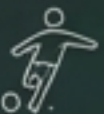
## MATCHES TABLE (12 Records)

| DATE  | STAGE    | HOME      | AWAY      | RES | STADIUM   |
|-------|----------|-----------|-----------|-----|-----------|
| 06-11 | Group    | Brazil    | Nigeria   | 3-1 | MetLife   |
| 06-12 | Group    | France    | USA       | 2-1 | AT&T      |
| 06-13 | Group    | Argentina | Mexico    | 4-2 | Rose Bowl |
| 06-14 | Group    | England   | Japan     | 2-2 | Azteca    |
| 06-15 | Group    | Germany   | Spain     | 1-1 | BC Place  |
| 06-20 | Group    | Brazil    | France    | 2-3 | SoFi      |
| 07-01 | Rd of 16 | England   | France    | 2-1 | Lumen     |
| 07-02 | Rd of 16 | Argentina | Germany   | 3-2 | NRG       |
| 07-06 | Q-Final  | Brazil    | Argentina | 1-2 | Mercedes  |
| 07-07 | Q-Final  | France    | England   | 2-1 | MetLife   |
| 07-10 | Semi     | Argentina | France    | 1-0 | AT&T      |
| 07-14 | FINAL    | Argentina | Brazil    | 2-1 | Rose Bowl |



# SAMPLE DATA | PERFORMANCE & FAN INSIGHTS

Granular in-match statistics and audience spending data



PLAYER\_PERFORMANCE (Sample of 25 Records)

| performance_id | match_id | player_id | minutes_played | goals_scored | assists | yellow_cards | red_cards | passes_completed | shots_on_target |
|----------------|----------|-----------|----------------|--------------|---------|--------------|-----------|------------------|-----------------|
| 1              | 1        | 1         | 90             | 2            | 1       | 0            | 0         | 45               | 4               |
| 2              | 1        | 2         | 75             | 1            | 0       | 1            | 0         | 38               | 3               |
| 3              | 1        | 17        | 90             | 0            | 0       | 0            | 0         | 52               | 0               |
| 4              | 2        | 3         | 90             | 1            | 1       | 0            | 0         | 52               | 5               |
| 5              | 2        | 12        | 85             | 1            | 0       | 0            | 0         | 41               | 2               |
| 6              | 3        | 5         | 90             | 2            | 1       | 1            | 0         | 67               | 6               |
| 7              | 3        | 6         | 80             | 2            | 0       | 0            | 0         | 35               | 4               |
| 8              | 3        | 18        | 90             | 0            | 0       | 0            | 0         | 28               | 0               |
| 9              | 4        | 7         | 90             | 1            | 1       | 1            | 0         | 28               | 3               |
| 10             | 4        | 8         | 88             | 1            | 0       | 0            | 0         | 55               | 2               |



FAN\_DEMOGRAPHICS (Sample of 15 Records)

| fan_id | match_id | age_group | gender | nationality | ticket_price_usd | merchandise_spent_usd |
|--------|----------|-----------|--------|-------------|------------------|-----------------------|
| 1      | 1        | 25-34     | Male   | Brazil      | 250.00           | 75.00                 |
| 2      | 1        | 35-44     | Female | USA         | 250.00           | 60.00                 |
| 3      | 2        | 25-34     | Male   | France      | 220.00           | 90.00                 |
| 4      | 2        | 18-24     | Male   | USA         | 220.00           | 45.00                 |
| 5      | 3        | 35-44     | Male   | Argentina   | 280.00           | 100.00                |
| 6      | 3        | 45-54     | Male   | Mexico      | 280.00           | 80.00                 |
| 7      | 4        | 25-34     | Female | England     | 300.00           | 70.00                 |
| 8      | 5        | 35-44     | Male   | Germany     | 200.00           | 55.00                 |
| 9      | 6        | 25-34     | Male   | Brazil      | 280.00           | 85.00                 |
| 10     | 7        | 18-24     | Female | USA         | 350.00           | 120.00                |



# SQL Analytics | Player & Team Performance

## CASE 1: TOURNAMENT TOP SCORERS ANALYSIS

Identifying high-performing athletes for media engagement

### BUSINESS SCENARIO

The broadcasting team wants to highlight top goal scorers for highlight reels.

### BUSINESS QUESTION

Who are the top 5 goal scorers and which teams do they represent?

### EXPECTED USE

Marketing team will use this for social media campaigns.

### QUERY SYNTAX

```
SELECT p.player_name, t.team_name,
       SUM(pp.goals_scorer) AS total_goals
FROM Player_Performance pp
JOIN Players p ON pp.player_id = p.player_id
JOIN Teams t ON p.team_id = t.team_id
GROUP BY p.player_name, t.team_name
ORDER BY total_goals DESC
LIMIT 5;
```

### TABLE QUERY RESULT

| RANK | PLAYER NAME    | TEAM      | TOTAL GOALS |
|------|----------------|-----------|-------------|
| 1 🏆  | Lionel Messi   | Argentina | 7           |
| 2 🥈  | Neymar Jr      | Brazil    | 5           |
| 3 🥉  | Kylian Mbappé  | France    | 4           |
| 4    | Harry Kane     | England   | 2           |
| 5    | Julian Alvarez | Argentina | 2           |

✓ Query executed successfully

0.03s execution time

## CASE 2: TEAM WIN-LOSS RECORD

Predicting tournament winners through performance analysis

### BUSINESS SCENARIO

The FIFA analytics team needs to determine which teams have the best performance record.

### BUSINESS QUESTION

Which teams have won the most matches in the tournament so far?

### EXPECTED USE

Used for power rankings and betting odds.

### QUERY SYNTAX

```
SELECT t.team_name, COUNT(*) AS total_wins
FROM Matches m
JOIN Teams t ON (m.team_home_id = t.team_id AND m.goals_home > m.goals_away)
              OR (m.team_away_id = t.team_id AND m.goals_away > m.goals_home)
GROUP BY t.team_name
ORDER BY total_wins DESC;
```

### TABLE QUERY RESULT

| RANK | TEAM NAME | TOTAL WINS |
|------|-----------|------------|
| 1 🏆  | Argentina | 5          |
| 2 🥈  | Brazil    | 2          |
| 3 🥉  | France    | 2          |
| 4    | England   | 1          |

✓ Query executed successfully

0.02s execution time



# SQL Analytics | Operational & Tactical Insights

## CASE 3: STADIUM UTILIZATION ANALYSIS

Optimizing venue operations and match scheduling



### BUSINESS SCENARIO

The FIFA operations team wants to understand which stadiums are attracting the most fans.



### BUSINESS QUESTION

What is the average attendance at each stadium and how many matches have been played there?



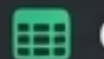
### EXPECTED USE

Stadium selection for high-profile matches.



### QUERY SYNTAX

```
SELECT s.stadium_name, s.city,
       AVG(m.attendance) AS avg_attendance,
       COUNT(m.match_id) AS total_matches
FROM Stadiums s
JOIN...
```



### QUERY RESULT

| STADIUM NAME    | CITY        | AVG ATTENDANCE | MATCHES |
|-----------------|-------------|----------------|---------|
| Rose Bowl       | Los Angeles | 86,750         | 2       |
| Azteca Stadium  | Mexico City | 87,000         | 1       |
| MetLife Stadium | New York    | 82,000         | 2       |
| AT&T Stadium    | Dallas      | 78,500         | 2       |



Query executed successfully  
0.05s execution time

## CASE 4: PLAYER PERFORMANCE EVALUATION

Assessing holistic player contributions for lineup decisions



### BUSINESS SCENARIO

National team coaches want to evaluate players' overall contributions.



### BUSINESS QUESTION

What are the comprehensive statistics for each player including goals, assists, and disciplinary records?



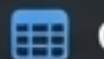
### EXPECTED USE

Team selection and tactical planning.



### QUERY SYNTAX

```
SELECT p.player_name, p.position,
       COUNT(pp.match_id) AS matches_played,
       SUM(pp.goals_scored) AS goals,
       SUM(pp.assists) AS assists,
       SUM(pp.yellow_cards) AS yellow_cards
FROM Players p JOIN...
```



### QUERY RESULT

| PLAYER NAME   | POSITION | MATCHES | GOALS | ASSISTS | Y.CARDS |
|---------------|----------|---------|-------|---------|---------|
| Lionel Messi  | Forward  | 5       | 7     | 2       | 2       |
| Neymar Jr     | Forward  | 4       | 5     | 2       | 0       |
| Kylian Mbappé | Forward  | 3       | 4     | 3       | 3       |



Query executed successfully  
0.04s execution time



# SQL Analytics | Financial & Fan Engagement

## CASE 5: REVENUE GENERATION BY TOURNAMENT STAGE

Financial analysis of ticket and merchandise sales across stages



### BUSINESS SCENARIO

The FIFA finance department needs to analyze revenue streams across different tournament stages.



### BUSINESS QUESTION

How much total revenue (tickets + merchandise) is generated at each stage?



### EXPECTED USE

Financial reporting and strategic budget allocation.



### QUERY SYNTAX

```
SELECT m.match_stage,
       COUNT(DISTINCT m.match_id) AS total_matches,
       SUM(fd.ticket_price_usd + fd.merchandise_spent_usd) AS
       total_revenue
FROM Matches m
JOIN...
```



### QUERY RESULT

| MATCH STAGE   | TOTAL MATCHES | TOTAL REVENUE (USD) |
|---------------|---------------|---------------------|
| Group         | 6             | \$2,885.00          |
| Final         | 1             | \$2,540.00          |
| Quarter-Final | 2             | \$1,630.00          |
| Semi-Final    | 1             | \$730.00            |
| Round of 16   | 1             | \$470.00            |

Query executed successfully. 0.05s execution time

## CASE 6: FAN DEMOGRAPHICS & SPENDING

Analyzing audience expenditure patterns for targeted marketing



### BUSINESS SCENARIO

The marketing team wants to identify which age demographics are spending the most.



### BUSINESS QUESTION

Which age groups spend the most on tickets and merchandise on average?



### EXPECTED USE

Targeted marketing campaigns and premium package design.



### QUERY SYNTAX

```
SELECT fd.age_group,
       COUNT(fd.fan_id) AS total_fans,
       AVG(fd.ticket_price_usd) AS avg_ticket_price,
       AVG(fd.merchandise_spent_usd) AS avg_merchandise
FROM Fan_Demographics fd
GROUP BY...
```



### QUERY RESULT

| AGE GROUP | TOTAL FANS | AVG TICKET (\$) | AVG MERCH (\$) |
|-----------|------------|-----------------|----------------|
| 35-44     | 6          | \$438.33        | \$115.83       |
| 25-34     | 6          | \$360.00        | \$115.00       |
| 45-54     | 2          | \$365.00        | \$115.00       |
| 18-24     | 2          | \$285.00        | \$82.50        |

Query executed successfully. 0.04s execution time



# SQL ANALYTICS | REPORTING & VALUATION

## CASE 7: MATCH RESULTS REPORT

Comprehensive match data integration for digital platforms

**BUSINESS SCENARIO**

The FIFA website needs a comprehensive match results page showing all completed matches with full details for fans and media.

**BUSINESS QUESTION**

What are all the match results with team names, scores, and venue details?

**EXPECTED USE**

Website content, mobile app updates, and press releases to keep fans informed.

**QUERY SYNTAX**

```
SELECT m.match_date, th.team_name AS home_team, ta.team_name AS away_team,
       m.goals_home, m.goals_away, m.match_stage, s.stadium_name
FROM Matches m
JOIN Teams th ON m.team_home_id = th.team_id
JOIN Teams ta ON m.team_away_id = ta.team_id
JOIN Stadiums s ON m.stadium_id = s.stadium_id
ORDER BY m.match_date;

-- Joins 3 tables to retrieve full context:
-- Home team, Away team, and Stadium details.
```

**LIVE DATA PREVIEW (TOP 5)**

| DATE       | HOME          | SCORE | AWAY        | STAGE | VENUE           |
|------------|---------------|-------|-------------|-------|-----------------|
| 2026-06-11 | BRA Brazil    | 3 - 1 | NGA Nigeria | Group | MetLife Stadium |
| 2026-06-12 | FRA France    | 2 - 1 | USA USA     | Group | AT&T Stadium    |
| 2026-06-13 | ARG Argentina | 4 - 2 | MEX Mexico  | Group | Rose Bowl       |
| 2026-06-14 | ENG England   | 2 - 2 | JPN Japan   | Group | Azteca Stadium  |
| 2026-06-15 | GER Germany   | 1 - 1 | ESP Spain   | Group | BC Place        |

✓ Data retrieved successfully | 12 records found

## CASE 8: PLAYER MARKET VALUE ANALYSIS

Identifying the most valuable assets in the tournament

**BUSINESS SCENARIO**

Football clubs and agents want to identify the most valuable players in the tournament for potential transfer negotiations and talent scouting.

**BUSINESS QUESTION**

Who are the top 3 most expensive players by market value in the tournament?

**EXPECTED USE**

Transfer market research, player scouting reports, and investment valuation for club owners.

**QUERY SYNTAX**

```
SELECT p.player_name, t.team_name, p.position, p.age, p.market_value_millions
FROM Players p
JOIN Teams t ON p.team_id = t.team_id
ORDER BY p.market_value_millions DESC
LIMIT 3;

-- Retrieves player valuation data sorted highest to lowest
-- to identify premium assets.
```

**QUERY RESULT**

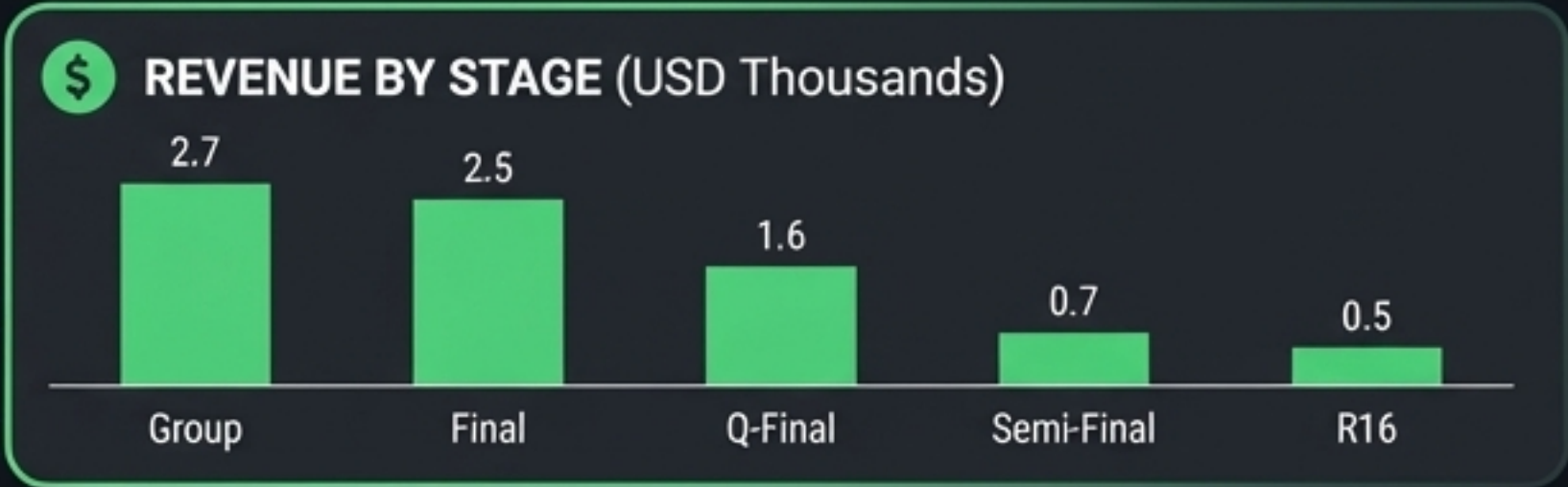
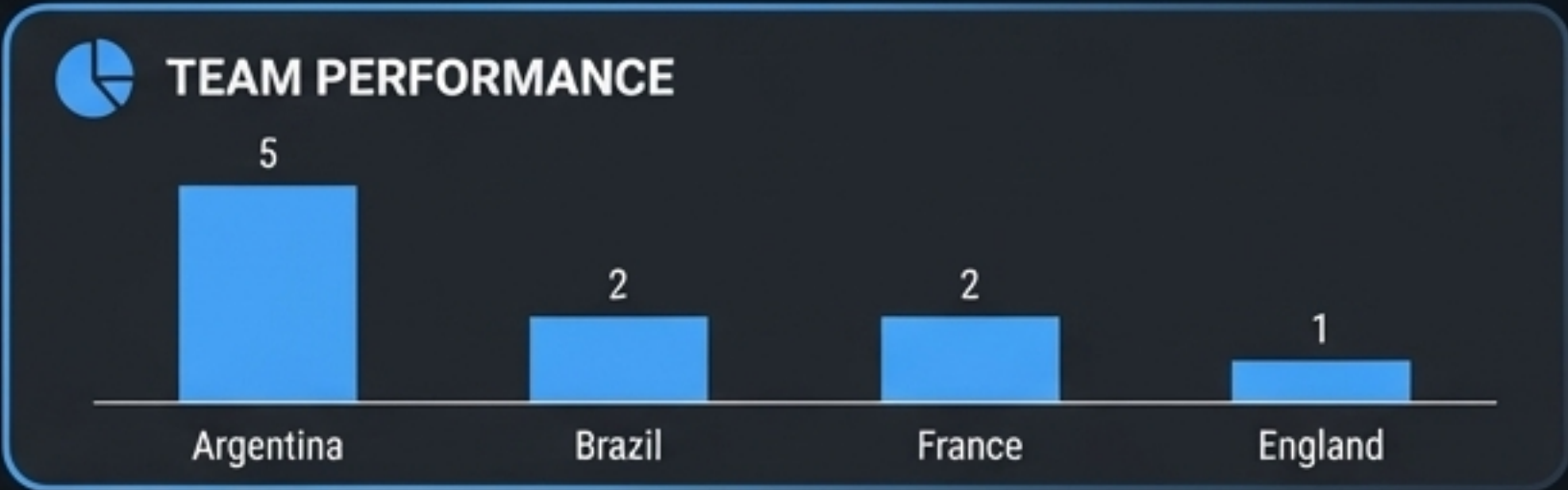
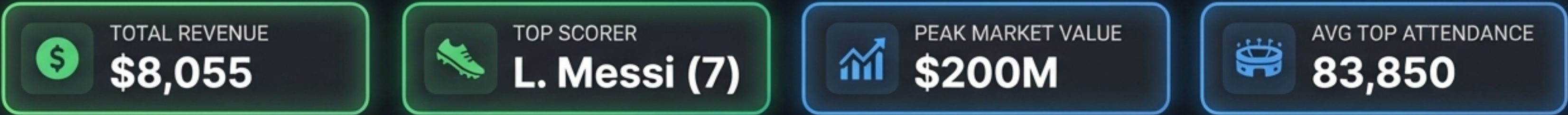
| RANK | PLAYER NAME     | TEAM    | POSITION   | AGE | MARKET VALUE (\$M) |
|------|-----------------|---------|------------|-----|--------------------|
| 1    | Erling Haaland  | Japan   | Forward    | 24  | 200.00             |
| 2    | Kylian Mbappé   | France  | Forward    | 25  | 180.00             |
| 3    | Jude Bellingham | England | Midfielder | 21  | 180.00             |

✓ Query executed successfully | 0.02s execution time



# THE COMMAND CENTER | UNIFIED TOURNAMENT DASHBOARD

Real-time analysis of player performance, revenue, fan demographics, and operations





**KEY INSIGHT:** The **35-44** age group exhibits the highest purchasing power for both tickets (\$438 avg) and merchandise, making them the primary target for premium hospitality packages.